

GCN-MHSA: A novel malicious traffic detection method based on graph convolutional neural network and multi-head self-attention mechanism

Jinfu Chen^{a,b}, Haodi Xie^{a,b}, Saihua Cai^{a,b,*}, Luo Song^{a,b}, Bo Geng^{a,b}, Wuhao Guo^c

^a School of Computer Science and Communication Engineering, Jiangsu University, 301 Xuefu Road, Zhenjiang, 212013, Jiangsu, China

^b Jiangsu Key Laboratory of Security Technology for Industrial Cyberspace, Jiangsu University, 301 Xuefu Road, Zhenjiang, 212013, Jiangsu, China

^c AsiaInfo Security Technology Co., Ltd Nanjing, 210012, Jiangsu, China

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ABSTRACT

With the increasing size and complexity of network, network traffic becomes more and more correlated with each other, and the traditional manner of presenting network traffic in a Euclidean structure is difficult to effectively capture the correlation information of network traffic. In contrast, graph structured data has gained much attention in recent years due to its ability to represent the correlation between different traffic flows; In addition, models and algorithms related to Graph Convolution Neural network (GCN) have been used for malicious traffic detection. However, existing GCN-based malicious traffic detection methods still suffer from incomplete description of the flow-level features of network traffic, imperfect traffic correlation establishment mechanism and failure to distinguish the importance of features during model training. Based on this, this study proposes a malicious traffic detection method called GCN-MHSA based on Graph Convolutional Neural network and Multi-Head Self-Attention mechanism. Firstly, the flow-level features of network traffic are populated and more information close to the features are selected to describe the network traffic; And then, the link homogeneity is used to establish the correlations between network traffic; Moreover, multi-head self-attention mechanism is introduced in the GCN model to provide larger weight to important features; Finally, an improved GCN is used as a deep learning model to detect malicious traffic. Extensive experimental results on three publicly available network traffic datasets and a real network traffic dataset show that the proposed GCN-MHSA method performs better than five baselines in terms of detection effect and stability, with an improvement of about 2.4% in accuracy, recall and F1-measure as well as an improvement of about 2.1% in precision.

1. Introduction

With the popularization of Internet, the scale of network traffic shows a continuous growth trend, and the complexity of network traffic is also increasing. Complex network architecture allows hackers to utilize more avenues for malicious attacks or performing abnormal behaviors, such as the communication in computer networks with attack behavior (Eswaran and Faloutsos, 2018) and the spread of false information in social networks (Gupta et al., 2012), which brings great challenges to network security. Compared with normal access, malicious attacks often generate malicious traffic with attack attributes, thus, we can identify the type of malicious attacks through detecting traffic and then take relevant measures to defend against network threats (Chen et al., 2023c). However, in real life scenarios, an access to a web page by users may trigger the download of multiple

resources and a response of a service may require multiple requests and responses, which leads to the strong correlation of network traffic reflecting users' operations. These correlated flows create complex interactions and dependencies in the network, therefore, understanding and analyzing such correlations can provide a more comprehensive view, which can help to more accurately detect malicious traffic. Graph structure can pay better attention on the correlation between network traffic due to its strong information expression ability, and some common dynamic networks, such as social networks, transaction networks for transferring money between accounts and computer communication networks, are all represented by graph-structured data. However, traditional malicious traffic detection methods based on recurrent neural network, convolutional neural network and their variants are not applicable to the detection task oriented to graph structure. Therefore,

* Corresponding author at: School of Computer Science and Communication Engineering, Jiangsu University, 301 Xuefu Road, Zhenjiang, 212013, Jiangsu, China.

E-mail addresses: jinfuchen@ujs.edu.cn (J. Chen), 2222108039@stmail.ujs.edu.cn (H. Xie), caisaih@ujs.edu.cn (S. Cai), 2212108063@stmail.ujs.edu.cn (L. Song), 2212308011@stmail.ujs.edu.cn (B. Geng), guo.wuhao@asiainfo-sec.com (W. Guo).

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how to build graph structures for the correlated network traffic and effectively detect malicious traffic with the help of graph structures that can better reflect the correlation of network traffic, has become a hot research topic.

In recent years, researchers tried to use graph neural network models to detect malicious traffic in large-scale network traffic. However, the existing researches on of network flow-level characterization still remains in the 5-tuple features of network traffic (including: source IP, source port, destination IP, destination port, protocol), which leads to the fact that these methods cannot fully reflect the complexity of network traffic characterization in the representation of traffic information (Sun et al., 2020). The existing methods also suffer from the problems of incomplete traffic correlation mechanism and not focusing on important features, which means that they may not be able to fully capture the correlations and interactions between different nodes in traffic data (Zheng et al., 2022). The incomplete correlation mechanism may cause the model ignoring some important information and patterns when analyzing network traffic, thus affecting its performance and accuracy. Moreover, existing methods fail to pay sufficient attention on important features when dealing with network traffic (Chen et al., 2023a). There is a large amount of information in network traffic data, but not all features are equally important to our study or task, while current methods may underutilize some key features due to not accurately evaluate and select the importance of features, thereby affecting the performance and accuracy of the model (Niu et al., 2022).

To address the problems of existing methods, we first propose a feature extending scheme to describe the network traffic as much detail as possible. More specifically, we extend the original network traffic features in the feature preprocessing stage and extract 12-tuple features of network traffic, where the source IP and port, the destination IP and port are used as two node features of in the graph structure, and other features such as the number of incoming and outgoing bytes, the number of incoming and outgoing packets, and the interval time are used as the edge features in the graph structure. In the constructing of graph structure for network traffic, we also consider the link homogeneity between network traffic (i.e., the similar network traffic may originate from same application) as well as further introduce the indirect correlation and direct correlation between network traffic to assign different weights to different degrees of correlation, thereby fully reflecting the correlation information between network traffic and take advantages of graph structure data. After constructing network traffic graph structure, we combine the Graph Convolutional Neural network and a Multi-Head Self-Attention mechanism to form an effective malicious traffic detection method called GCN-MHSA, thereby further improving the stability of detection model and the detection efficiency of malicious network traffic.

The main contributions of this paper are as follows:

1. We extend the traditional 5-tuple flow-level features of network traffic into 12-tuples to describe the network traffic with richer information.
2. We introduce the indirect and direct correlations between network traffic using link homogeneity between network traffic and assign different weights to different degrees of correlation, thereby more effectively describing the correlations between network traffic.
3. We introduce the multi-head self-attention mechanism into graph convolutional neural network to provide larger weight for more important features, thereby improving the stability and detection efficiency of malicious network traffic method.

The remainder of this paper is organized as follows. Section 2 presents the related work on malicious traffic detection. Section 3 describes the proposed GCN-MHSA method, including the preprocessing of network traffic, the establishment of graph structure, and the construction of GCN-MHSA method. Section 4 makes an experimental comparison between the proposed GCN-MHSA method and state-of-the-art methods. Finally, Section 5 concludes this paper and discusses the future work.

2. Related work

Malicious traffic detection aims to detect abnormal network traffic from large-scale network traffic to seek for the attack behavior or unusual operations, it is an important aspect of network security. With the development of artificial intelligence, machine learning, deep learning and graph neural network techniques have been widely used in malicious traffic detection.

2.1. Machine learning-based malicious detection methods

Machine learning-based malicious detection methods distinguish malicious traffic from normal traffic by learning the features of network traffic using machine learning models. For example, Moore and Zuev (2005) proposed a Bayesian classifier-based malicious detection method, it had high detection accuracy, but the time consumption used in feature extraction process is very high. Zhang et al. (2018) proposed a network intrusion detection method that combines Principal Component Analysis (PCA) and Gaussian naive Bayes to detect malicious traffic, it weighted the first few features obtained by traditional PCA to reduce the data dimension, thereby reducing the time consumption. Alshammari and Zincir-Heywood (2009) conducted a large number of experiments and found that using C4.5 classification can improve the detection accuracy and robustness on SSH and Skype networks compared with Adabost, Naive Bayes and RIPPER, but this approach produced a high false positive rate. Okada et al. (2011) proposed an EFM method based on SVM to detect malicious traffic from encrypted traffic with an overall accuracy of 97.2%, but this method did not consider the optimal combination of features for network traffic. Shams and Rizaner (2018) introduced a network traffic intrusion detection tool using support vector machine, which significantly improved network reliability by detecting and removing malicious nodes in the network. In 2016, Anderson and McGrew (2016) proposed a supervised machine learning model, it combined context information and observable data in the network traffic to detect malicious traffic. Wu et al. (2021) preprocessed network packets by collecting network traffic data, system logs, user behavior and host information, and then used random forest algorithm to train the data; In addition, they also calculated the importance of features to reduce data dimensionality. Barradas et al. (2021) developed the Flowlens, it relied on programmable switches to collect flow distribution information on the data-plane for feature extraction and detected the attacks by applying a random forest algorithm. However, this method did not consider the external relationships between traffic flows, which may limit its effectiveness in capturing complex network behaviors and interactions. Holland et al. (2021) proposed nPrint, a tool that generated a unified packet representation by using the maximum header length allowed for each protocol and concatenating different protocol headers into a fixed-length header, it integrated the nPrint with automated machine learning (AutoML) for automatically classifying network traffic. However, nPrint offered only a little assistance in extracting flow-level sequence information, which limited its effectiveness in feature extraction of network traffic. In recent years, several studies have made significant contributions to the field of real-time malicious traffic detection. For example, Fu et al. (2021) developed the Whisper leveraging frequency domain features, it realized the robust and real-time detection in high throughput network. Zhou et al. (2023) proposed a multi-phase sequential model architecture, this method introduced a new paradigm for the joint design and deployment of intelligent network models and programmable network hardware. Fu et al. (2023) proposed the HyperVision, an unsupervised graph learning method, which used the DBSCAN and KD-Tree to analyze the connectivity, sparsity and statistical features of the graph for flow interaction patterns, thereby detecting the unknown patterns in encrypted malicious traffic. Compared with these methods, the GCN-based methods require the construction of extensive graph structures and entail higher computational overhead, they offer a distinct

advantage by capturing the complex dependencies and relationships between different network traffic, leading to more effective detection of malicious traffic.

In general, machine learning-based malicious traffic detection methods attempt to discriminate normal network traffic and malicious network traffic using machine learning models. Although this kind of malicious traffic detection methods has made some progress, they suffer from the following problems: (1) The feature extraction process is time-consuming in large-scale networks and the dimensionality reduction techniques do not always preserve all essential information, leading to the loss of potential information; (2) Most methods face high false positive rate due to the need of feature extraction with greater subjectivity.

2.2. Deep learning-based malicious detection methods

Compared with traditional machine learning-based malicious detection methods, deep learning-based models have the advantage of directly extracting features from original network traffic, thus, it has been widely used in the detection of malicious traffic in recent years. For example, Prasse et al. (2017) proposed a long short-term memory (LSTM)-based model to detect malware based on the Https traffic with the focus on host address, timestamp and data information. Wang et al. (2017a) proposed an automatic malware detection method, it used the natural language processing to extract text features from network traffic and then used the text features to construct an effective malware detection model based on text semantic features. Wang et al. (2017b) proposed a malicious traffic detection method based on one-dimensional convolutional neural network (1D-CNN), but this method only used spatial features and ignored temporal features. Marín et al. (2019) used raw measurements of monitored byte streams as input to the proposed model and evaluated different raw traffic representation to better capture the basic statistics of malicious traffic. In addition, Chen et al. (2023b) proposed a malicious traffic detection method based on temporal convolutional network (TCN), it used exponential linear units (ELU) activation function to solve the problem of low detection accuracy caused by the neuron “death” in the training process. Sun et al. (2020) developed an intrusion detection system called DL-IDS with the use of hybrid network (containing CNN and LSTM) to extract the spatial and temporal features of network traffic, but its detection accuracy for some attack types needs to be improved. Wu et al. (2020) designed a LDoS attack classifier using HSMM network model as well as proposed a LDoS attack detection method based on wavelet energy spectrum entropy and hidden semi-Markov model, they had better detection performance. Mirsky et al. (2018) developed the Kitsune to learn the per-packet features by adopting auto-encoders, it was an unsupervised method and could eliminate the need for labeled datasets as well as was capable of handling zero-day attacks. However, this packet-level method used the statistics as the context information, resulting in unable to achieve robust detection under evasion attacks.

In general, deep learning-based methods provide an effective way to detect malicious network traffic, they also have better accuracy and portability compared with traditional machine learning-based methods. However, deep learning-based methods primarily focus on extracting spatial and temporal features from network traffic, these methods face the challenges in effectively capturing the intricate network relationships and graph structures associated with malicious activities.

2.3. Graph neural network-based malicious detection methods

Considering the advantages of graph structure in representing network traffic, researchers have begun to study graph neural network (GNN)-based models and algorithms in recent years. In GNN-based malicious traffic detection methods, researchers ensure the detection reliability through different data preprocessing methods, feature extraction

schemes and model optimization. For example, Kipf and Welling (2017) proposed a scalable semi-supervised GNN-based learning method, it directly acted on GNN structure based on an effective variant of CNN to effectively improve the detection efficiency. Fang et al. (2022) proposed a switching aware spatio-temporal GNN model based on dynamic graph convolution and gated linear unit to predict the traffic consumption in short, medium and long time frames, but this method could not obtain a hybrid graph structure with detailed information about network traffic. After a lot of analysis, Zola et al. (2022) found that the key to malicious traffic detection is not only to detect a single malicious connection, but also to detect which node in the network is generating malicious traffic in many cases, and then they tool corresponding measures to reduce the threat and improve the network security of system; However, this method had a bad detection efficiency problem due to the lack of denoising processing on network traffic. Xiao et al. (2023) proposed a control area network-based graph attention network (CAN-GAT) model to realize anomaly detection for in-vehicle network, and then proposed a malicious detection framework based on graph convolution network and graph attention network to detect anomalies in CAN bus in-vehicle network, but the detection stability of this method needs to be further verified. Guo et al. (2020) proposed a GNN-based anomaly detection algorithm, it introduced graph structure, attribute and dynamic change information into the model to learn the representation vector of anomaly detection, but the operation efficiency of this method needs to be improved. Zheng et al. (2022) proposed an encrypted malicious traffic detection method called GCN-ETA based on graph convolutional network as well as a new traffic representation and correlation construction scheme, but the traffic direction and the weights of different correlation types were not considered in the traffic correlation. Lan et al. (2022) proposed E-minBatch, a model based on GraphSAGE, it took the application layer source port and source IP address as the source node, the application layer target port and target IP address as the target node, and the remaining traffic information as the edge information, thereby completing the construction of graph structure, but this method was limited to the detection of industrial Internet traffic. Jiang et al. (2022) designed a new trust evaluation framework called GATrust based on GAT, it could assign different attention coefficients to various attributes of users in online social networks and improve the prediction accuracy of social trust evaluation; However, the generality of this method on graph structure data needs to be verified. Niu et al. (2022) proposed a method based on adaptive online analysis to accurately determine malware via analyzing the encrypted, drifting and unbalanced network traffic. The GNN-based model Euler (King and Huang, 2022) belonged to unsupervised method, it utilized the temporal graph link prediction to incorporate the temporal nature of data along with GNN model for anomaly-based intrusion detection. The unsupervised learning method Euler exhibited higher false positive rate and false negative rate in some cases, it was owing to that unsupervised methods rely on the inherent structure of the data without the guidance of supervised signals, making them more susceptible to noise compared to supervised methods. (1) In the feature extraction, the edges in the graph structure of Euler model did not include any information of network traffic, they only represented the existence of relationships between two entity nodes, leading to the loss of much useful information. (2) Regarding the correlation between network traffic, Euler model treated all correlations between network traffic uniformly and fails to account for the varying degrees of relevance and importance among different network traffic, it may hinder the effectiveness of malicious activities detection. (3) In the training of the detection model, Euler had the advantage of not requiring labeled data, it outperformed other unsupervised anomaly-based intrusion detection methods; However, Euler model assigned equal weights to each feature, this manner could significantly limit its ability to capture the importance of different features.

In summary, the studies on GNN-based malicious detection have become a hot research topic in recent years. However, the existing

Table 1
The comparison of malicious traffic detection methods.

Method	Feature extraction	Using graph structure	Considering feature weights
Zhang et al. (2018)	Processed	×	×
Shams and Rizaner (2018)	Processed	×	×
Wang et al. (2017a)	Direct	×	×
Wang et al. (2017b)	Direct	×	×
Marín et al. (2019)	Direct	×	×
Zheng et al. (2022)	Direct	✓	×
Lan et al. (2022)	Direct	✓	×
Jiang et al. (2022)	Direct	✓	×
Niu et al. (2022)	Processed	×	×
Sun et al. (2020)	Direct	×	×
GCN-MHSA (ours)	Direct	✓	✓

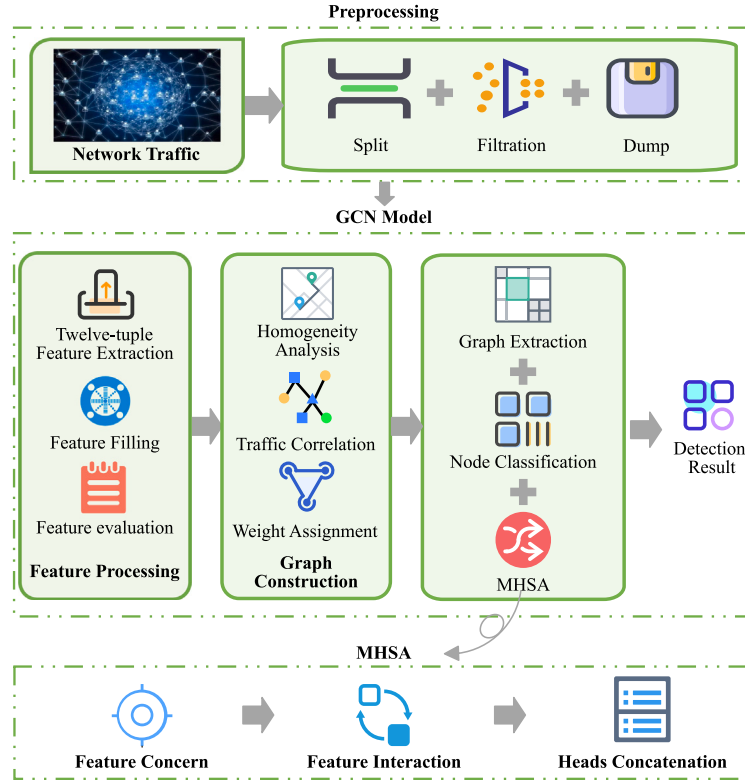


Fig. 1. The workflow of GCN-MHSA method.

researches do not give a complete set of flow-level features to describe network traffic and ignore the link homogeneity between network traffic when establishing their correlation, which seriously affect the efficiency of GNN-based malicious traffic detection methods. Therefore, it is necessary to strengthen the preprocessing process of original network traffic, improve the attention to the correlation between network traffic, enhance the effectiveness of GNN-based model in the training phase and strengthen the attention on important features of network traffic, thereby further improving the detection efficiency of malicious traffic.

The comparison of some aforementioned malicious traffic detection methods is shown in Table 1.

3. Proposed model

To solve the problems of existing GCN-based malicious traffic detection methods, such as the incomplete description of flow-level features of network traffic, the incomplete traffic correlation mechanism and not focusing on important features in the model training process, we propose a malicious traffic detection method called GCN-MHSA based on graph convolution neural network and multi-head self-attention

mechanism. The overall workflow of proposed GCN-MHSA method is shown in Fig. 1.

Firstly, the basic preprocessing operations such as data segmentation, filtering and dumping are performed on the original network traffic. Then, the features of processed network traffic are extracted and extended with the flow-level features, and the selected features are evaluated to ensure that the selected features have positive feedback. Next, the correlation between network traffic is established using link homogeneity to assign different degrees for indirect and direct correlations. Finally, different features are assigned different weights through multi-head self-attention mechanism based on their importance to the determination of malicious traffic. With above operations, the adjacency matrix, flow-level features and other information are provided for the training of malicious traffic detection model to more accurately detect malicious traffic. Compared with existing methods, the proposed method extends the flow-level features of network traffic, that is, selecting more information close to the features of network traffic to describe it, as well as analyze the homogeneity between network traffic, and then introduces the multi-headed self-attention mechanism into GCN model to assign different weights for features. Through these improvements, the effect of malicious traffic detection is

improved while fully considering the relevant information and features between network traffic.

3.1. The preprocessing of network traffic based on feature extending and network correlation analysis

The GCN-based malicious traffic detection method can use the traffic trajectory graph to construct graph structure. Generally, the nodes in the graph structure are represented by the IP and port, and the edges in the graph structure represent the information of network traffic. From the perspective of logical reasoning, the node information in the graph structure is IP and port and the edge information in the graph structure is the related network protocol number, that is, the flow-level feature tuple of network flow is $F(src_IP, dst_IP, src_P, dst_P, Pro)$.

Although this practice considers the advantages of using graph structure to represent network traffic, but the contained information is far from covering the specific characteristics of a network traffic. In order to take full advantage of graph structure to store network traffic, the feature information to be stored on each network traffic is first extended with the features. Specifically, because the IP address and port of each network traffic are sufficient to describe each node, and too much description of endpoint information of each network traffic may not only cause data redundancy, but also affect the construction efficiency of graph structure, thus, the information of features contained in each node of graph structure is not changed. For the description information on the edges of graph structure, the protocol number information used by most researchers to describe the features of network traffic is obviously insufficient, where the protocol number usually only provides rough information about the protocol to which the traffic belongs but lacks more specific information about the details and behavior of traffic camouflage, which will have a negative effect on the detection efficiency of malicious traffic. For this reason, this study not chooses to use protocol number to describe the network traffic, but selects some features most relevant to each network flow, including the number of inflow bytes ($num(in_b)$), the number of outflow bytes ($num(out_b)$), the sum number of bytes ($num(sum_b)$), the length of inflow packets ($len(in_p)$), the length of outflow packets ($len(out_p)$), the sum number of packets ($num(sum_p)$), the interval time between current network traffic and last network traffic ($int(time)$), and the duration of current network traffic ($dur(nt)$). The above key features of network traffic are extracted by the tshark (a Wireshark's built-in command-line tool). This process aims to build a more comprehensive and accurate set of features to support subsequent tasks of network traffic analysis and malicious traffic detection. These features are described as follows:

1. $num(in_b)$, $num(out_b)$, $num(sum_b)$: The number of inflow bytes of malicious traffic ($num(in_b)$), the number of outflow bytes of malicious traffic ($num(out_b)$) and the sum number of inflow and outflow bytes of malicious traffic ($num(sum_b)$) may indicate potential malicious behavior or intrusion attempts, thus, the malicious traffic patterns of network attacks or other abnormal conditions can be detected by counting the number of bytes flowing in and out, thereby detecting and preventing the network security threats.
2. $len(in_p)$, $len(out_p)$, $num(sum_p)$: Different types of applications or protocols usually have different distributions of packet length and different sum number of packet, these features can be used to identify and distinguish different types of network traffic, such as Web browsing, file transfer or video streaming. In addition, the packet length also can be used to analyze the delay characteristics of network traffic. Therefore, the malicious traffic patterns of network attacks or other abnormal conditions can be detected by counting the length and total number of packets.

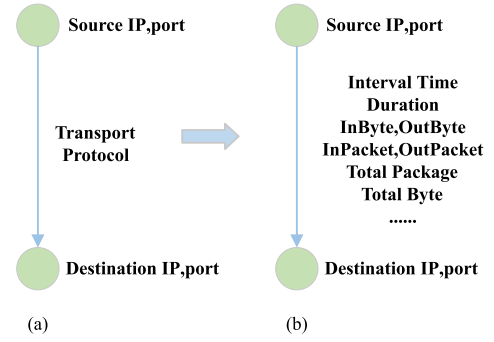


Fig. 2. The extended schematic of flow-level features.

3. $int(time)$, $dur(nt)$: Different types of applications or protocols usually have different traffic time interval and different duration, observing the interval time between current network traffic and last network traffic as well as duration of current network traffic can reveal regular, periodic or abnormal traffic events, which can help to distinguish and identify specific applications, user behaviors or network events.

With the above considered factors, the extended 12-tuple is $F(src_IP, dst_IP, src_P, dst_P, in_b, out_b, in_p, out_p, sum_p, sum_b, int, dur)$, where $src_IP: src_P$ and $dst_IP: dst_P$ are the direct relationship between each IP and its port, $sum_p = in_p + out_p$, $sum_b = in_b + out_b$. The extended schematic of flow-level features is shown in Fig. 2, where Fig. 2(a) is the features of network traffic before extending and Fig. 2(b) is the features of network traffic after extending. This study uses these information of features to describe each network traffic in more detail, so as to improve the detection efficiency of malicious network traffic.

3.2. The correlation conversion of network traffic based on link homogeneity and weight assignment

The network traffic with same IP host are more likely to be the same software compared with randomly selected network traffic, that is, there is link homogeneity in the network (Zheng et al., 2022) (link homogeneity refers to the degree of similarity between network correlations and transmission paths in the network). Taking whether there is a common IP between two network traffic as the basis for judging whether there is a correlation between network traffic, the edge classification in network traffic detection can be transformed into a node classification problem. The edge classification problem is usually more complex and requires more time overhead because it needs to consider every edge in the network. Compared with edge classification problem, the node classification focuses on the classification at the node level, and it is easier to obtain label because nodes are usually directly related to specific identifiers (such as IP address or hostname). Considering that network traffic has flow direction, this study improves the original undirected graph of traffic correlation into a directed graph and further defines the direct and indirect correlation between network traffic, as well as assigns different weights to the direct and indirect correlation to consider more important correlated information. The improved structure of graph conversion is shown in Fig. 3.

The definition of graph in Fig. 3 is $G = (V, A)$, where V is the set of nodes and A is a symmetric adjacency matrix. The elements a_{ij} of adjacency matrix are the weights of edges between nodes v_i and v_j , where $a_{ij} = 0$ indicates that there is no edge correlation between two nodes, $a_{ij} = 1$ indicates that there is an indirect correlation between two edges, and $a_{ij} = 2$ indicates that there is a direct correlation between two edges. In addition, we also define the degree matrix $D = \text{diag}$

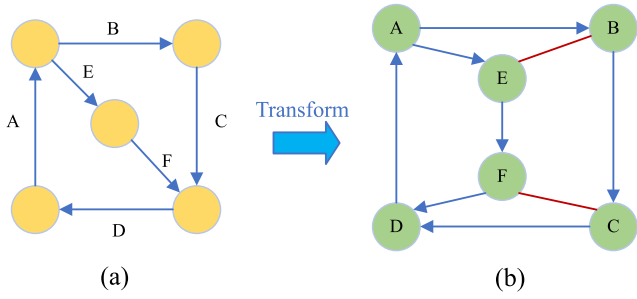


Fig. 3. The correlation conversion of network traffic.

$(d_1, \dots, d_n)$, where each node v_i in the graph has a corresponding d -dimensional feature vector. The feature vectors of all n nodes form a complete feature matrix X .

According to the above definition, suppose there are n network flows and each has d -dimensional features, feature matrix X and adjacency matrix A . Specifically, when two network traffic share the same IP host, a_{ij} can take the value of 1 or 2, respectively, which represents the degree of correlation between network traffic. If there is no IP host sharing, a_{ij} is set to 0, indicating that there is no correlation between network traffic. In addition, in order to better distinguish the importance of different correlations, we define the cases in which there is a correlation between two network traffic:

1. Direct correlation: When the source IP address of one traffic is the same as the destination IP address of another network, these two network traffic are regarded as direct correlation, where the value of corresponding position is set to 2 in the adjacency matrix, indicating that they are more connected, and this higher weight helps to highlight the importance of direct connected network traffic.
2. Indirect correlation: When two network traffic has same source IP address or destination IP address, these two network traffic are regarded as indirectly connected, where the value of corresponding position is set to 1 in the adjacency matrix. This lower weight indicates that although they may be connected with same IP address, they may not have close communication, therefore, their relevance is not as high as that of direct connected network traffic.

This strategy of different weight allocation can better distinguish the importance of different correlations. Typically, direct connected network traffic imply the closer communication and they should be assigned larger weights; While indirect correlations imply that although such correlations are still considered in the connected network traffic, they may involve the same IP address but do not necessarily indicate direct communication, they should be assigned smaller weights. This strategy helps to more accurately capture the correlation information between network traffic and reflect the degree of these correlations in the adjacency matrix.

As shown in Fig. 3, the destination IP of traffic A is the source IP of traffic B, so there is a direct correlation between them, thus, the value of its corresponding position in the adjacency matrix is set to 2. Traffic B and E have the same source IP, and traffic C and F have the same destination IP, that is, B and E, C and F are indirectly connected, thus, the value of their corresponding position in the adjacency matrix is set to 1.

3.3. The framework and algorithm of GCN-MHSA

After constructing graph structure for network traffic, we use an improved GCN model called GCN-MHSA to detect malicious traffic. Traditional GCN-based methods can learn the correlated features between

nodes, but it is difficult to model the complex dependency between the nodes in network traffic. To solve this problem, we introduce multi-head self-attention mechanism and correlation matrix into the GCN model to assign different weights to direct correlation and indirect correlation. Among them, the correlation matrix allows the weights of edges in the graph to be adjusted to more accurately reflect the degree of dependency between nodes, which can help GCN to aggregate more accurate features of network traffic nodes. The overall framework of GCN-MHSA model is shown in Fig. 4.

GCN is a multi-layer neural network model that provides a powerful tool for malicious traffic detection. It can directly process the data in graph structure and use the attributes of adjacent nodes to guide the embedding vector learning of each node, thereby obtaining the feature representations of new node. These feature representations are further used as input to a linear classifier for the classification task of nodes. In addition, through the stacking of multi-layer GCN, a wider range of information can be gradually integrated to effectively analyze the relationship between network topology and node attributes as well as identify abnormal behavior, thereby more comprehensively capturing the relevant information in the detection of malicious traffic. For the k th graph convolution layer, the matrices $H^{(k-1)}$ and $H^{(k)}$ represent the representations of input node and output node, respectively. In general, the representation of initial node is the original input feature $H^{(0)} = X$, which represents the input of first graph convolution layer. For a single-layer GCN, the new d' -dimensional node feature matrix $H^{(1)}$ is shown in formula (1).

$$H^{(1)} = \text{ReLU}(MX\theta^{(1)}) \quad (1)$$

In formula (1), the normalized adjacency matrix M of self-loop is added, that is, the connection between nodes and themselves in the graph is also considered when the adjacency matrix is normalized, which is shown in formula (2).

$$M = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} \quad (2)$$

In formula (2), $\tilde{A} = A + I$, $\tilde{D}$ is a degree matrix, the weight matrix $\theta^{(1)}$ used to smooth the linear transformation of the hidden feature representation. In the GCN model, multiple GCN layers can be stacked to capture higher-order domain information, which is shown in formula (3).

$$H^{(k)} = \text{ReLU}(MH^{(k-1)}\theta^{(k)}) \quad (3)$$

For node classification, the last layer of GCN uses classifier to predict labels. The predicted class of n nodes is defined as $\hat{Y} = \hat{Y}_{ic} n \times c$, where $\hat{Y}_{ic}$ represents the probability that node i belongs to class C . Therefore, the e-class prediction of GCN at layer K is shown in formula (4).

$$\hat{Y}_{GCN} = \text{softmax}(MH^{(K-1)}\theta^{(K)}) \quad (4)$$

Assuming constructing a two-layer GCN ($K = 2$), the overall forward propagation mode is shown in formula (5).

$$\hat{Y} = f(X, A) = \text{softmax}(M \times \text{ReLU}(MX\theta^{(1)})\theta^{(2)}) \quad (5)$$

Finally, the cross-entropy loss function of all labeled nodes is calculated, and the sequence of features in high-dimensional network traffic is obtained by inputting the traffic features of graph structure into GCN.

In view of the complex inter-correlations between network traffic, it is urgent to deeply focus on and clearly express the complex relationships between different network traffic. Therefore, we introduce a multi-head self-attention mechanism into GCN model, it uses multiple different attention heads to calculate the relationships between nodes, thereby improving the expressive ability of the model. In particular, it is worth noting that the application of multi-head self-attention mechanism in the GCN-MHSA hidden layer means that the self-attention mechanism is introduced between the node representations of each layer, thereby accurately capturing the dependency between nodes.

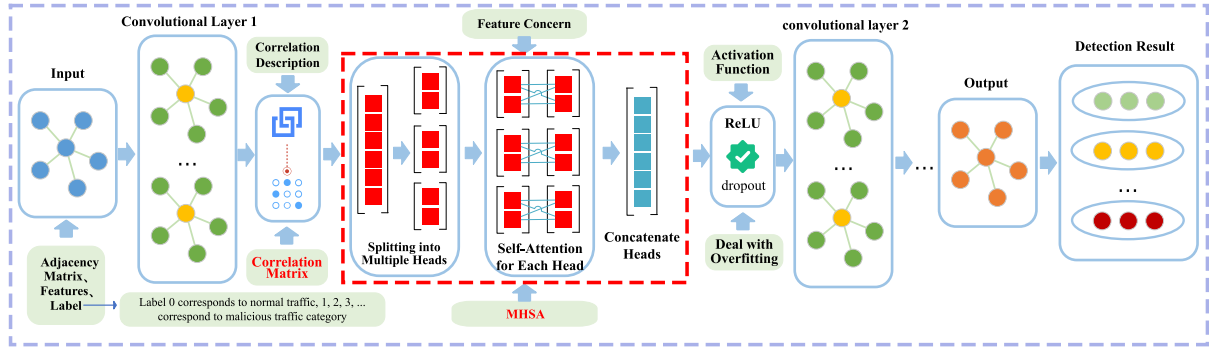


Fig. 4. The framework of GCN-MHSA model.

This improvement permits a more detailed analysis and understanding of the correlations between different network traffic, thus further improving the accuracy and performance of the model detection.

Assuming that the input feature matrix is $X_{n \times d}$, where n represents the number of traffic sequences and d represents the feature dimension of each traffic sequence. Firstly, the multi-head attention mechanism is used to weight the input features and produce an attention matrix $A_{n \times n}$, as shown in formula (6).

$$A = \text{softmax}\left(\frac{XW^Q(XW^K)^T}{\sqrt{d_k}}\right) \quad (6)$$

In formula (6), W^Q and W^K are matrices used to project input feature X onto query matrix Q and key matrix K , and d_k represents the dimension of each head. Multi-head attention mechanism uses h different W^Q and W^K to learn the dependency between nodes, resulting in h attention matrices $A_1, A_2, \dots, A_h$. Softmax operation normalizes the weights in the attention matrix to range $[0,1]$, $\sqrt{d_k}$ is used for scaling to prevent excessive inner product being too large. The calculation of attention matrix for each head is shown in formula (7).

$$\text{head}_i = \text{softmax}\left(\frac{XW_i^Q(XW_i^K)^T}{\sqrt{d_k}}\right), i = 1, \dots, h \quad (7)$$

For each attention matrix head_i , we can use the same way as the traditional GCN to calculate its feature representation Z_i , which is shown in formula (8).

$$Z_i = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A}_i \tilde{D}^{-\frac{1}{2}} X W_i^V) \quad (8)$$

In formula (8), Z_i is obtained by encoding the feature X of the i th head node, $\tilde{A}_i$ is the attention matrix with self-loop added, $\tilde{D}$ is the diagonal matrix, and the weight matrix W_i^V is used to perform linear transformation on X and project the input feature onto the value V . $d \times v$ is the output dimension of each head. The multi-head feature representation $Z_{d \times v}$ can be obtained by connecting the feature representation of each head, which is shown in formula (9).

$$Z = \text{concat}(Z_1, \dots, Z_h) \quad (9)$$

Finally, a fully connected layer is used to fuse the multi-head feature representation Z , which is shown in formula (10).

$$H^{l+1} = \text{ReLU}(\tilde{D}^{-\frac{1}{2}} Z W^{\text{out}}) \quad (10)$$

In formula (10), W^{out} is the weight matrix used to project the multi-head feature representation to the next hidden layer, and H^{l+1} is the feature representation of the next hidden layer.

The detailed process of malicious traffic detecting by GCN-MHSA model is shown in Algorithm 1. Firstly, the parameters including learning rate, number of epoch and dropout rate are set to initialize the model; And then, the data in graph structure is input into the GCN-MHSA model for processing. Within each training epoch, iterate through each batch of data to extract features from graph structure,

and then obtain labels based on classification results of GCN model as well as calculate the loss and update the weight matrices to obtain a high-dimensional set of correlated features after each training batch. Subsequently, apply the multi-head self-attention mechanism to process the correlated features and focus on the most important features. Finally, the detection operation is performed to obtain the malicious traffic.

Algorithm 1 GCN-MHSA

Input: $G(V, A)$, learning rate, epoch, dropout, batch

Output: Detection result

```

1: SetParameters(learningrate, epoch, dropout)
2: LoadData( $G$ )
3:  $G \rightarrow$  GCN-MHSA model
4: for each epoch do
5:   for each epoch do
6:     ExtractFeatures()
7:     classification result based on GCN  $\rightarrow$  label
8:     CalculateLoss()
9:     renew weight values  $W$ 
10:    obtain the association feature  $AF$ 
11:   end for
12:    $AF = \text{MultiHeadAttention}(AF)$ 
13: end for
14:  $Result = \text{Classification}(AF)$ 
15: return Result

```

The proposed GCN-MHSA model can achieve more powerful and flexible feature extraction through multi-head self-attention mechanism, multi-head feature representation and multi-head feature fusion. In the multi-head self-attention mechanism, each head focuses on different features, which provides the model with multiple attention methods to better capture complex patterns in network traffic. In addition, multi-head feature representation and multi-head feature fusion enhance the feature representation and help to characterize network traffic more comprehensively. This manner also allows weighted aggregation of nodes' neighboring nodes, which further enriches the representation of nodes and enables the model to capture the complex relationships among nodes more accurately.

Through flexibly combining the attention weights of different heads, the model can more accurately express the relationship between nodes, which improves the generalization ability of the model and enables it better adapt to different network traffic datasets and tasks. In addition, the introduction of multi-head self-attention mechanism helps to reduce the impact of noise on the model for improving the robustness of the model as well as significantly improves the learning efficiency and training convergence speed of the model, thereby accelerating the training process of the model, improving the detection performance and better adapting to the changing network environment.

Table 2

The description of UJS-IDS2022 dataset.

Normal	Filesize (KB)	Malware	Filesize (KB)
Tencent video	32 450	ArkeiStealer	159 477
QQ music	30 220	AsyncRAT	27 492
Thunder	492 530	AveMariaRAT	297 535
Youku	288 026	Formbook	379 488
Iqiyi	40 592	Hive	560 345
Mango video	179 003	Kutaki	234 207
Himalaya	8944	Loki	17 261
Baidu post bar	89 530	MassLogger	171 313
League of legends	13 096	OskiStealer	77 368
Bilibili	108	RedLineStealer	160 061

4. Experiment

This section first describes the dataset, evaluation metrics, experimental settings and baselines used in the experiment, and then verifies the effectiveness of the proposed GCN-MHSA method with five state-of-the-art methods and discusses the experimental results. Subsequently, the ablation experiment is conducted to verify the effectiveness of the improved scheme proposed in this study.

4.1. Dataset description

CIC-IDS2017 dataset is a collection of network attack data collected by the Canadian Cyber Security Research Institute from computer simulated network attack scenarios on multiple local area networks, it is produced and maintained by researchers from the University of New Brunswick, Canada. This dataset is generated from a large-scale simulation of network traffic, including both normal network traffic and malicious network traffic. Among them, malicious network traffic includes 16 different types of network attack, such as DoS, Scan, R2L, U2R and others.

CTU-13 is a dataset for cybersecurity research, it is provided by the researchers from Computer Science Department of Czech Technical University. The dataset contains network traffic captured in a real network environment, which contains normal network traffic as well as samples from 13 different types of attacks, including: worms, Trojans, DDoS attacks, botnets, sniffing, scanning, malware downloads, etc. Each attack sample contains multiple network sessions, and each session has a set of packets and associated metadata information, such as timestamp, IP address, port number and so on.

USTC-TFC2016 dataset is created and collected by the University of Science and Technology of China (USTC) to analyze and detect malicious traffic, it contains multiple types of malicious traffic. The images in the dataset come from different scenes and environments in the real world and cover multiple categories of objects to make it having diversity and representative.

UJS-IDS2022 dataset is the collected network traffic produced by multiple normal software and malware in the real environment by the Key Laboratory of Industrial Cyberspace Security Technology of Jiangsu University from June 5, 2022 to June 18, 2022. The number of network traffic in this dataset is 7.67 million (2.76 million for benign, 4.91 million for malicious). The specific software and the size of PCAP files corresponding to their network traffic are shown in Table 2.

4.2. Evaluation metrics

In the experiments, four evaluation metrics are selected to evaluate the detection efficiency of proposed GCN-MHSA model for malicious network traffic, they are shown in Table 3.

In Table 3, TP (True Positive) refers to the number of malicious traffic correctly identified by the GCN-MHSA method as malicious traffic; FP (False Positive) refers to the number of normal traffic incorrectly identified by the GCN-MHSA method as malicious traffic; TN (True

Negative) refers to the number of normal traffic correctly identified by the GCN-MHSA method as normal traffic; FN (False Negative) refers to the number of malicious traffic incorrectly identified by the GCN-MHSA method as normal traffic.

4.3. Experimental setup

The hardware environment of the experiment is two 20-core 40-thread CPUs, 128G RAM and two RTX3090 GPUs, and the software environment is Ubuntu 20.04.2 LTS x86_64, TensorFlow 2.4.0 and Python 3.8.

To determine the optimal model parameters, we employ grid search to systematically explore a wide range of configurations. The parameters includes the batch_size of 16, 32 and 64; the dropout rate of 0.4, 0.5 and 0.6; the epoch_size of 10, 30, 50 and 70; and the learning rate of 0.1, 0.01 and 0.001. Each parameter combination is evaluated for 30 times and the hyperparameter under highest detection accuracy is selected as the parameter of our GCN-MHSA model, including batch_size = 64, dropout rate = 0.5, epoch_size = 50, and learning rate = 0.01. To ensure the fairness across experiments, all baselines are trained using identical hyperparameter settings. Additionally, after several experimental iterations, we incorporate three layers of multi-head self-attention mechanism with four attention heads in the model to balance the performance with computational cost. The ratio of training datasets to testing datasets is 8:2 for all used four network traffic datasets.

4.4. Baselines

In order to verify the validity of the proposed GCN-MHSA model, five advanced and representative baselines are used in the experiment for comparison.

1. E-minBatch GraphSAGE (Lan et al., 2022). It is an improved GraphSAGE neural network model that can adapt to complex network environment.
2. GATrust (Jiang et al., 2022). It is an improved graph attention network that can assign different attention coefficients in the network.
3. IARF (Niu et al., 2022). It is an improved encryption malicious traffic detection method based on adaptive random forest.
4. CNN-LSTM (Sun et al., 2020). It is a hybrid network of CNN and LSTM that can effectively capture and extract the spatio-temporal features of network traffic.
5. GCN-ETA (Zheng et al., 2022). It is an encrypted malicious traffic detection method based on graph convolutional neural networks.

4.5. Experimental results and analysis

4.5.1. The performance analysis of multiclassification

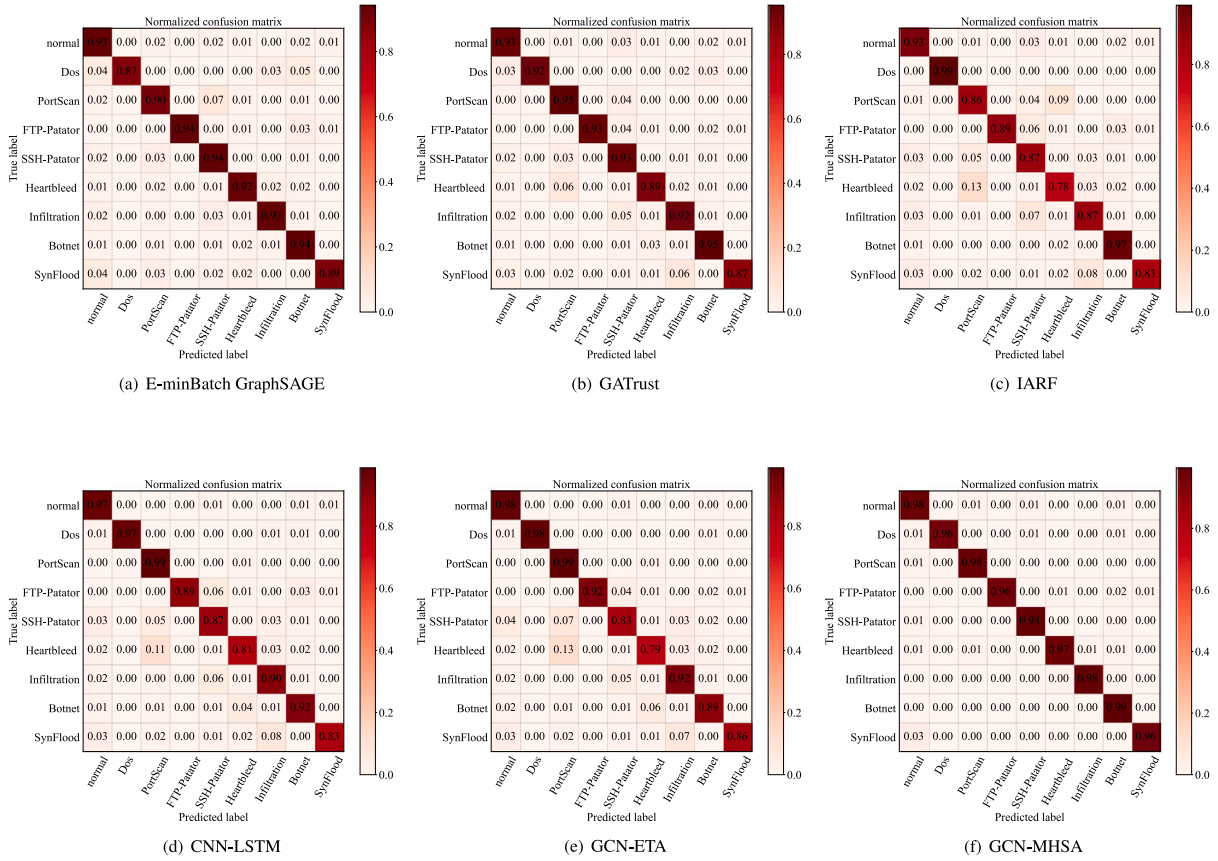
In order to evaluate the multi-classification performance of the proposed GCN-MHSA method, we conduct a large number of experiments on three public network traffic datasets and a real network traffic dataset with the comparison of five baselines. Fig. 5 shows the experimental results on CICIDS-2017 dataset using confusion matrix, where the horizontal axis represents the class predicted by the GCN-MHSA method and the vertical axis represents the true class of network traffic. The values in the matrix have been normalized, and the diagonal elements correspond to the detection accuracy of the model.

As is shown in Fig. 5(f), GCN-MHSA method has good performance in malicious traffic detection with an accuracy close to 98%. Fig. 5(a) shows that the E-minBatch GraphSAGE method has a poor detection efficiency on certain types of malicious traffic when processing a small number of samples, and its overall performance is lower than that of proposed GCN-MHSA method due to it is prone to overfitting in the

Table 3

Evaluation metric.

Metric	Formula	Meaning
Precision	$P = \frac{TP}{TP+FP}$	The proportion of samples predicted to be positive that are actually positive.
Accuracy	$A = \frac{TP+TN}{TP+FP+TN+FN}$	The proportion of samples whose predictions are correct.
Recall	$R = \frac{TP}{TP+FN}$	The proportion of positive samples that are predicted to be positive.
F1-measure	$F1 = \frac{2 \times R \times P}{R+P}$	The accuracy rate and recall rate are considered comprehensively, it is the harmonic average of the two metrics.

**Fig. 5.** The confusion matrix of six compared malicious traffic detection methods.

learning process. The experimental results in Fig. 5(b) and (c) show that the GATrust and IARF methods exhibit poor detection performance compared to the GCN-MHSA model in identifying network traffic with certain types of attack (such as Dos, Heartbleed, synflood, PortScan and SSH-Patator), which indicates the poor generalization ability of these two methods. As is shown in Fig. 5(d) that the CNN-LSTM method also has some shortcomings similar to the IARF method. Nevertheless, the CNN-LSTM method shows comparable detection accuracy to the GCN-MHSA method in identifying normal network traffic as well as the malicious traffic with the attack types of Dos and PortScan. Fig. 5(e) shows that although the detection efficiency of GCN-ETA method on Dos and PortScan attacks is close to that of GCN-MHSA, but its detection efficiency on other types of attack is not good.

The experimental results show that traditional malicious traffic methods that do not include feature weight learning can successfully learn most of the correlated features in both normal traffic and large-scale malicious traffic. However, they are not efficient in detecting malicious traffic in small sample categories. In contrast, the proposed GCN-MHSA method introduces a multi-head self-attention mechanism, which can weight each element in the input sequence according to the correlation of different positions in the input sequence, thus obtaining accurate and reliable detection results.

4.5.2. The performance of GCN-MHSA

To verify whether the proposed GCN-MHSA method can obtain better detection capability of malicious traffic, we compare it with five methods, including E-minBatch GraphSAGE, GATrust, IARF, CNN-LSTM, and GCN-ETA. Each experiment is conducted for 30 times and the average experimental result is shown in Table 4, where the positive and negative errors are the standard deviations of 30 repetitions.

The experimental results on CICIDS-2017 dataset show that the proposed GCN-MHSA method has better performance in malicious traffic detection than E-minBatch GraphSAGE, GATrust, IARF, CNN-LSTM and GCN-ETA methods. Specifically, compared with E-minBatch GraphSAGE method, the accuracy, precision, recall and F1-measure of GCN-MHSA method improve by 2.15%, 2.05%, 1.89% and 2.31%, respectively; Compared with GATrust, these evaluation metrics improve by 1.92%, 1.46%, 1.48% and 1.62%, respectively; Compared with IARF, the accuracy, precision, recall and F1-measure metrics of GCN-MHSA method increase by 2.6%, 2.24%, 2.5% and 2.77%, respectively; Compared with CNN-LSTM, these metrics improve by 1.26%, 1.14%, 2.2% and 2.0%, respectively; Compared with GCN-ETA, the above metrics increase by 4.3%, 3.9%, 4.36% and 4.42%, respectively. Similarly, a similar effect is also can be observed on the CTU-13 dataset and USTC-TFC2016 dataset.

In addition, in order to verify the detection efficiency of the proposed GCN-MHSA method in a real data environment, we conduct a

Table 4
The comparison of six malicious traffic detection methods.

Methods	Dataset	Accuracy (%)	Precision (%)	Recall (%)	F1-measure (%)
E-minBatch GraphSAGE	CICIDS-2017	95.86 $\pm$ 0.35	96.02 $\pm$ 0.77	96.32 $\pm$ 1.64	95.83 $\pm$ 0.84
	CTU-13	96.33 $\pm$ 0.73	96.76 $\pm$ 0.89	95.18 $\pm$ 1.52	96.34 $\pm$ 0.52
	USTC-TFC2016	97.55 $\pm$ 0.66	96.98 $\pm$ 0.91	97.76 $\pm$ 0.82	97.33 $\pm$ 0.69
	UJS-IDS2022	91.36 $\pm$ 1.71	91.82 $\pm$ 1.56	92.55 $\pm$ 0.49	92.18 $\pm$ 1.32
GATrust	CICIDS-2017	96.11 $\pm$ 0.87	96.61 $\pm$ 1.42	96.73 $\pm$ 0.53	96.52 $\pm$ 0.48
	CTU-13	97.67 $\pm$ 1.02	97.77 $\pm$ 1.36	97.41 $\pm$ 0.76	97.25 $\pm$ 0.93
	USTC-TFC2016	98.22 $\pm$ 0.54	98.56 $\pm$ 0.75	97.93 $\pm$ 0.42	98.20 $\pm$ 0.77
	UJS-IDS2022	96.28 $\pm$ 1.02	96.96 $\pm$ 0.98	97.16 $\pm$ 1.29	97.06 $\pm$ 0.67
IARF	CICIDS-2017	95.43 $\pm$ 0.92	95.83 $\pm$ 0.96	95.71 $\pm$ 0.96	95.37 $\pm$ 0.86
	CTU-13	96.13 $\pm$ 0.85	96.23 $\pm$ 0.62	95.43 $\pm$ 0.65	95.43 $\pm$ 0.79
	USTC-TFC2016	96.81 $\pm$ 0.93	97.04 $\pm$ 0.73	96.57 $\pm$ 0.88	96.82 $\pm$ 1.07
	UJS-IDS2022	92.66 $\pm$ 1.02	92.49 $\pm$ 0.89	93.19 $\pm$ 0.98	92.84 $\pm$ 0.91
CNN-LSTM	CICIDS-2017	96.77 $\pm$ 1.32	96.93 $\pm$ 0.92	96.01 $\pm$ 0.72	96.11 $\pm$ 0.39
	CTU-13	97.12 $\pm$ 1.63	97.37 $\pm$ 0.63	96.96 $\pm$ 0.34	97.04 $\pm$ 0.81
	USTC-TFC2016	97.98 $\pm$ 0.92	98.48 $\pm$ 0.25	98.18 $\pm$ 0.32	98.39 $\pm$ 0.63
	UJS-IDS2022	95.67 $\pm$ 2.03	95.96 $\pm$ 1.75	96.18 $\pm$ 1.44	96.07 $\pm$ 1.69
GCN-ETA	CICIDS-2017	93.73 $\pm$ 0.57	94.17 $\pm$ 0.89	93.85 $\pm$ 1.00	93.85 $\pm$ 1.00
	CTU-13	95.72 $\pm$ 0.74	94.90 $\pm$ 1.12	95.47 $\pm$ 0.57	95.18 $\pm$ 0.35
	USTC-TFC2016	97.48 $\pm$ 0.21	97.57 $\pm$ 0.64	96.44 $\pm$ 0.71	96.99 $\pm$ 0.45
	UJS-IDS2022	90.51 $\pm$ 1.45	90.78 $\pm$ 1.22	91.37 $\pm$ 0.89	91.07 $\pm$ 1.01
GCN-MHSA	CICIDS-2017	98.03 $\pm$ 0.15	98.07 $\pm$ 0.15	98.21 $\pm$ 0.23	98.14 $\pm$ 0.08
	CTU-13	99.06 $\pm$ 0.19	98.76 $\pm$ 0.13	99.03 $\pm$ 0.18	98.90 $\pm$ 0.06
	USTC-TFC2016	99.31 $\pm$ 0.13	99.03 $\pm$ 0.21	99.47 $\pm$ 0.23	99.12 $\pm$ 0.19
	UJS-IDS2022	96.38 $\pm$ 0.36	96.87 $\pm$ 0.27	97.18 $\pm$ 0.42	97.02 $\pm$ 0.34

series of experiments on the UJS-IDS2022 dataset. The experimental results show that compared with the experiments on public network traffic, the detection efficiency in the real environment are generally lower because of the diversity and complexity of collected network traffic. Nevertheless, the proposed GCN-MHSA method still achieves the best results in the real environment. It is worth noting that GCN-MHSA performs slightly worse than the GATrust method on the evaluation metrics of precision and F1-measure, it is possibly due to the GATrust method has the ability of feature representation with strong learning adaptability, which results in it adapting to different topologies and sizes of network traffic. Compared with the state-of-the-arts, the proposed GCN-MHSA method has greater stability, which shows that GCN-MHSA is an excellent detection method in the real environment, especially in the face of diversity and complexity of network traffic.

Compared with other methods, the proposed GCN-MHSA method achieves excellent experimental results on multiple evaluation metrics due to the following advantages: (1) In the model preprocessing stage, through considering and utilizing the information embedded in the network traffic, we extend the features of network traffic to make the model capturing the overall structure and attributes of network traffic more comprehensively as well as avoiding information loss or bias caused by missing data; (2) We introduce the multi-head self-attention mechanism to capture different types of interactions between nodes more comprehensively and focus on important features, thereby making the model able to learn different attention weights in parallel to assign different weights for the features in network traffic; (3) Through modeling and analyzing network topology, GCN-MHSA model can extract valuable information and patterns from complex network traffic, this graph structure-based manner enables GCN-MHSA method to capture the relationship and interaction between nodes effectively, which is benefit it for learning the hidden features in network traffic. In the compared methods, GATrust performs best because it considers more relationships among nodes in the process of constructing graph structure through feature transformation. The CNN-LSTM hybrid network has the third highest detection performance on all datasets by extracting the spatiotemporal features of network traffic. In contrast, E-minBatch GraphSAGE model only uses the information of local neighbor node but not uses the global information, that is, it does not make full use of the overall graph structure information, which results in its detection performance weaker than GATrust and

CNN-LSTM. In addition, the detection performance of IARF model is weaker than other models except GCN-ETA because it only learns the features from online streaming data, which limits its ability to learn from large-scale datasets. Compared with above compared methods, the detection results of GCN-ETA are weak, it is owing to that it does not assign different weights to different levels of traffic correlations, so unimportant features produce a greater impact on its detection results.

4.5.3. The stability of GCN-MHSA

To verify the stability of the proposed GCN-MHSA method, we conduct the experiments on four datasets for 30 times with all compared methods, and the boxplot according to the experimental results is shown in Fig. 6.

As can be seen from Fig. 6 that compared with E-minBatch GraphSAGE, GATrust, IARF, CNN-LSTM and GCN-ETA methods, the proposed GCN-MHSA method has significant performance advantages and higher stability. The GCN-MHSA model always show higher accuracy, precision, recall and F1-measure, and no outliers are observed in the experimental results. It is worth to note from Fig. 6(a) to (d) that the interquartile range (indicating by the box length) is significantly narrower for the GCN-MHSA model, indicating a higher stability of its detection efficiency. The higher stability of GCN-MHSA method is owing to that it extends the flow-level features of network traffic and optimizes the correlation algorithm of network traffic; In addition, it also use the multi-head self-attention mechanism to focus on those more important features. In contrast, the compared methods exhibit a wider range of fluctuations and have some outliers. The experimental results on stability show that compared with five state-of-the-art malicious detection models, the proposed GCN-MHSA method has higher accuracy and more stable detection effect.

4.5.4. Ablation experiment

To verify the influence of each module on the proposed GCN-MHSA model, we perform a series of ablation experiments, each experiment is conducted for 30 times and the average experimental result is shown in Table 5, where the positive and negative errors are the standard deviations of 30 repetitions. Among them, GCN is the original graph convolutional neural network model, it uses 5-tuples of features for detection. GCN-FF adds the feature extending technology on the basis of GCN model in the preprocessing stage, and it uses extended 12-tuples

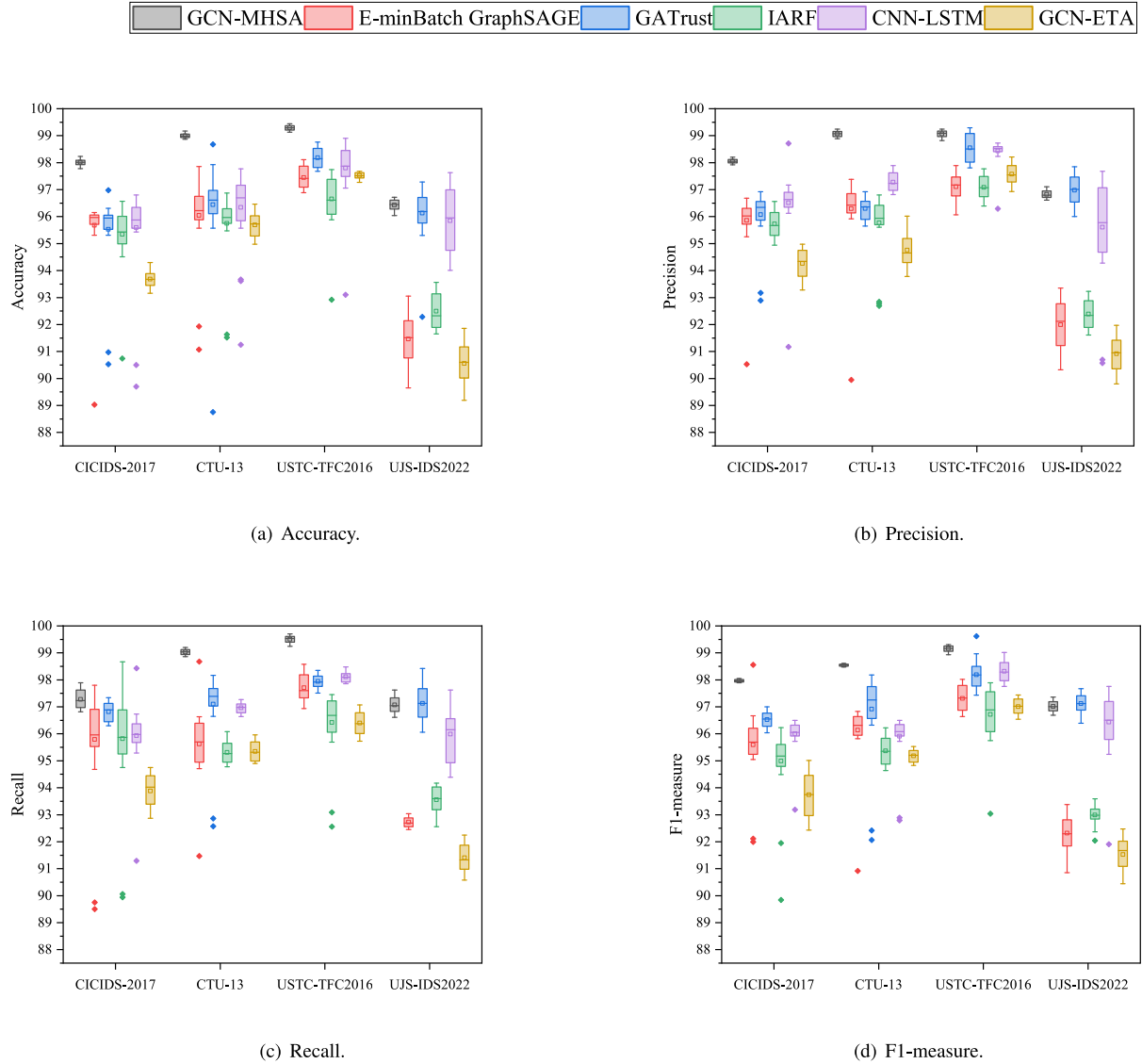


Fig. 6. The stability of six compared malicious detection methods.

of features for detection. GCN-CO optimizes the mechanism of traffic correlation establishment on the basis of GCN model, that is, setting the indirect correlation and direct correlation, and uses 5-tuples of features for detection. GCN-M introduces a multi-head self-attention mechanism on the basis of GCN model, it uses 5-tuples of features for detection. GCN-FF-CO adds feature extending module and optimizes traffic correlation mechanism on the basis of GCN-FF. GCN-FF-M introduces the multi-head self-attention mechanism based on the GCN-FF model. GCN-CO-M introduces a multi-head self-attention mechanism based on the GCN-CO model, and it uses 5-tuples of features for detection.

It can be seen from the results of ablation experiments in Table 5 that removing any module will lead to a decrease in the detection efficiency of malicious traffic, and the multi-head self-attention mechanism plays the most critical role among these modules. After the introduction of multi-head self-attention mechanism, the detection accuracy on all datasets has been significantly improved, which is due to the fact that multi-head self-attention mechanism can adaptively learn the correlation and importance between features, which is benefit for the model to flexibly assign different weights for the features according to task requirements, and thus making the model has good universality ability. In addition, the feature extending and optimizing traffic correlation establishment mechanism in the preprocessing stage also play

Table 5

The influence of different modules on detection accuracy of GCN-MHSA.

Model	CICIDS-2017 (%)	CTU-13 (%)	USTC-TFC2016 (%)	UJSIDS2022 (%)
GCN	91.47 ± 1.71	94.81 ± 1.05	97.96 ± 0.97	93.11 ± 1.98
GCN-FF	92.09 ± 1.67	95.29 ± 1.23	98.33 ± 0.85	93.76 ± 1.34
GCN-CO	91.93 ± 1.55	95.04 ± 1.47	98.17 ± 1.01	94.03 ± 1.61
GCN-M	95.72 ± 1.21	98.19 ± 1.57	99.08 ± 1.01	95.43 ± 1.33
GCN-FF-CO	95.37 ± 1.12	95.53 ± 1.08	98.43 ± 0.46	94.51 ± 1.40
GCN-FF-M	97.84 ± 0.17	99.01 ± 0.24	99.19 ± 0.15	96.23 ± 0.68
GCN-CO-M	97.91 ± 0.24	98.45 ± 0.41	94.14 ± 0.17	94.03 ± 0.52
GCN-MHSA	98.03 ± 0.15	99.06 ± 0.19	99.31 ± 0.13	96.38 ± 0.36

a significant role in improving the detection accuracy, especially on the CICIDS-2017 dataset. This is because the dataset contains many attack types and provides detailed attack labels, which results in the optimization of these modules can get better experimental results. It can be seen from Table 5 that the proposed GCN-MHSA method achieves the best detection efficiency and has better generalization on all four datasets, which also shows that these improvements in the GCN-MHSA method have a good effect on improving the detection efficiency of malicious traffic.

Table 6
The time cost of six malicious traffic detection methods (s).

Model	Time cost	USTC-TFC2016	CTU-13	Stratosphere
E-minBatch GraphSAGE	Training process	236.33	323.57	217.53
	Detecting process	4.48	7.99	7.37
	Overall	240.81	331.56	224.90
GATrust	Training process	231.32	361.55	317.31
	Detecting process	7.76	9.01	7.31
	Overall	239.08	370.56	324.62
IARF	Training process	282.51	272.56	262.06
	Detecting process	12.33	17.51	9.98
	Overall	294.84	290.07	272.04
CNN-LSTM	Training process	137.13	148.65	123.01
	Detecting process	5.37	5.09	7.32
	Overall	142.50	153.74	130.33
GCN-ETA	Training process	166.41	173.38	153.33
	Detecting process	8.53	11.71	8.87
	Overall	174.94	185.09	162.20
GCN-MHSA	Training process	172.49	170.20	161.51
	Detecting process	9.01	7.88	7.35
	Overall	181.50	178.08	168.86

4.5.5. Time efficiency of GCN-MHSA

In addition to detection accuracy, the time efficiency is another important factor to measure the efficiency of malicious traffic detection method. To verify the time efficiency, we conduct extensive experiments on three widely used public network traffic datasets, and the experimental result is shown in Table 6.

It can be seen from the experimental results in Table 6 that the time cost of GCN-MHSA is only slightly higher than that of CNN-LSTM and GCN-ETA, it consumes less time than other graph neural network-based malicious traffic detection. For the CNN-LSTM model, it achieves the highest time efficiency because it is not a graph neural network, thus inherently faster in processing; However, CNN-LSTM neglects the inter-node relationships, resulting in the lower detection accuracy. The reason for consuming shorter time of GCN-ETA is that it uses an identity matrix to replace the weight matrix, which simplifies the computations; However, this simplification decreases the classification accuracy.

The superior time efficiency of our model compared to other models can be largely attributed to the incorporation of multi-head self-attention (MHSA) mechanism, which enhances efficiency of GCN-MHSA model through two key aspects. (1) MHSA enables simultaneous attention to multiple parts of the graph, which allows the model to capture diverse node relationships and interactions comprehensively. This distributed focus facilitates more efficient learning by ensuring that the model integrates a broader range of graph features during each training iteration. With the use of MHSA, GCN-MHSA model can converge more quickly because it gathers richer and more nuanced information from the graph, enhancing its ability to learn complex patterns and relationships. (2) The parallel processing capability of MHSA optimizes the computational efficiency by enabling simultaneous processing of different parts of the graph, it means that the model can handle diverse graph features concurrently, which potentially leads to faster training times as well as more effective using of computational resources, this advantage is particularly beneficial for large-scale datasets and complex graph structures.

4.5.6. The application of GCN-MHSA

GCN-MHSA model is designed for deployment at the network gateway. GCN-MHSA detects the malicious traffic through analyzing the structured data of network traffic at the gateway. Firstly, GCN effectively captures the relationships and dependencies between different network entities, such as IP addresses, ports and protocols, and then identifies the patterns indicative of malicious behavior. And then, MHSA enhances the model's ability to focus on various aspects of network traffic simultaneously. At the gateway, the proposed GCN-MHSA model can consider multiple features concurrently, such as the

number and the size of packets, the interval time, the duration of current network traffic, etc. (that is, the twelve-tuple features used in GCN-MHSA), thereby detecting the malicious activities with high accuracy.

GCN-MHSA model is designed to detect a variety types of malicious traffic, including but not limited to: (1) Brute Force FTP: Attacks targeting FTP services by attempting numerous username and password combinations to gain unauthorized access. (2) Brute Force SSH: It is similar to FTP brute force attacks, but the targeting is SSH login credentials. (3) DoS: Attacks that overload the target system with a flood of requests, rendering it unable to provide normal services. (4) Heartbleed: Exploits vulnerabilities in OpenSSL to read protected memory, leading to the leakage of sensitive information. (5) Web Attacks: Attacks against web applications such as SQL injection and cross-site scripting (XSS), aiming to gain the unauthorized access or corrupt data integrity. (6) Infiltration: Attacks gain the initial access and subsequently penetrate deeper into the network to access more resources or cause damage. (7) Botnet: Networks of compromised devices coordinated to perform attacks such as DDoS or spam distribution. (8) DDoS: Large-scale attacks using multiple controlled devices to overwhelm and incapacitate a target system.

Although the proposed GCN-MHSA model is not specifically designed against evasion attacks, it can possess the robust defenses against evasion tactics due to it combines the global feature extraction, multi-head self-attention mechanism, dynamic adaptability and the stability of graph structures. In the GCN-MHSA model, GCN is adept at capturing the global structural features of network traffic by analyzing the overall graph structure, which makes it challenging for attackers to evade detection by merely altering specific features of network traffic; In addition, the stability and robustness of graph structures make it difficult for attackers to significantly alter the overall network traffic structure without raising suspicion. The MHSA mechanism enhances the model's sensitivity and robustness by capturing the features of network traffic from multiple perspectives and scales. Even if attackers attempt minor modifications to evade detection, the multi-head mechanism aggregates the information from various attention heads to find potential anomalies. Furthermore, the self-attention mechanism is dynamically adaptive, it adjusts the weights of features to capture the evolving information of network traffic, this adaptability can ensure that the detection model maintains high detection capabilities in the face of new or mutated attacks.

5. Conclusion

In this paper, we propose a malicious traffic detection method called GCN-MHSA based on graph convolutional neural network and multi-head self-attention mechanism. Firstly, the information contained in

the network traffic is enriched by extending the flow-level features. And then, the algorithm of establishing traffic correlation relationship is optimized to better capture the features of malicious traffic by accurately establishing the correlation relationship between traffic, thereby making the model to identify potential threats more accurately and detect malicious traffic effectively. Subsequently, the multi-head self-attention mechanism is integrated into graph convolutional neural network model to effectively capture the dependence between nodes and makes full use of local neighborhood information of nodes through network structure, it also promotes the generation of multiple feature combinations through stacked self-attention layers. With the use of attention mechanism, the feature combinations that are more important for malicious traffic detection are assigned larger weight to improve the detection efficiency for malicious traffic. The proposed GCN-MHSA method provides an accurate and efficient method for malicious traffic detection by fusing graph convolutional neural network and multi-head self-attention mechanism and optimizing the traffic correlation construction algorithm, which is expected to play an important role in the field of network security. Extensive experiments on four network traffic show that the proposed GCN-MHSA method performs better than E-minBatch GraphSAGE, GATrust, IARF, CNN-LSTM and GCN-ETA in terms of accuracy, precision, recall and F1-measure, and it also has better stability.

Although the proposed GCN-MHSA method achieves better detection results, it also has some shortcomings to be further improved in the future:

1. Although current flow-level features of network traffic have been greatly expanded than original five-tuples, the process of extending flow-level features is a complex and long process, it is necessary to study a more comprehensive and reliable framework to extract more effective flow-level features to further improve the detection efficiency of malicious traffic.
2. In the multi-head self-attention model, it is possible to further optimize the weight learning ability of features. In the future, it can be considered to introduce more levels of attention structure to make the model can select and integrate more fine-grained features, thereby further improving the model's ability to learn the correlation and importance between different features.

CRedit authorship contribution statement

Jinfu Chen: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Data curation. **Haodi Xie:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation. **Saihua Cai:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Data curation. **Luo Song:** Writing – review & editing, Data curation. **Bo Geng:** Writing – review & editing. **Wuhao Guo:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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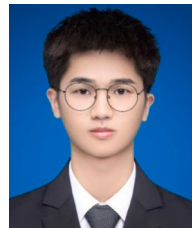
Jinfu Chen received the Ph.D. degree in computer science and technology from Huazhong University of Science and Technology, Wuhan, China, in 2009. He is currently a full professor in the School of Computer Science and Communication Engineering, Jiangsu University, Zhenjiang, China. His major research interests include software testing, software security, and trusted software. He has published more than 80 papers in some famous journals or conferences, including in *ACM Transactions on Privacy and Security*, *IEEE Transactions on Reliability*, *Information Sciences*, *Journal of Systems and Software*, *Information and Software Technology*, *Software: Practice and Experience*, *IET Software*, *The Computer Journal*, *ISSTA*, *ASE*, *ISSRE*, and *QRS*. He is a member of the IEEE and the ACM, and a member of the China Computer Federation.



Haodi Xie received the B.E. degree in 2021 in Software Engineering from Nantong Institute of Technology, Nantong, China. He is currently working towards the Master's degree in Computer Technology at Jiangsu University, Zhenjiang, China. His research interest includes malicious traffic detection and graph neural network.



Saihua Cai received his Ph.D. degree in Agricultural Engineering from China Agricultural University, Beijing, China, in 2020. He is currently an associate professor in the School of Computer Science and Communication Engineering, Jiangsu University, Zhenjiang, China. His major research interests include malicious traffic detection, outlier detection and software testing. He has published more than 70 papers in journals or conferences, including in *ACM Transactions on Privacy and Security*, *IEEE Transactions on Reliability*, *Information Sciences*, *Computers & Security*, *The Computer Journal*, *Knowledge-Based Systems*, *ISSRE*, and *QRS*, etc. He is a member of the IEEE and the ACM, and a member of the China Computer Federation.



Luo Song received the B.E. degree in 2021 in Software Engineering from Jiangsu University, Zhenjiang, China. He is currently working towards the Master's degree in Computer Technology at Jiangsu University, Zhenjiang, China. His research interest includes encrypted malicious traffic detection.



Bo Geng received the B.E. degree in 2020 in Material science and technology from Harbin institute of technology, Harbin, China. He is currently working towards the Master's degree in Control Science and Engineering at Jiangsu University, Zhenjiang, China. His research interest includes malicious traffic detection and adversarial attacks and defenses.



Wuhao Guo received the Master degree from Nanjing University, he is currently working towards the Ph.D degree at the School of Cyberspace Security, Southeast University. He has joined in AsiaInfo Security in 2021, holding positions as the Vice President, General Manager of the COO Office, General Manager of the Solutions and Technical Support Department, and General Manager of the Digital Support Department. His major research interests include malicious traffic detection, data security, and software security.